

Beta Adjustment

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1 Introduction

The standard Beta estimator for a stock calculates Beta as the slope coefficient in a linear regression of the returns on the individual stock against a market index, typically taken to be the S&P 500. The time period covered by the returns usually spans daily observations over a two- to five-year period. What makes the estimation process especially tricky is that Beta itself may change over time as the firm's financial characteristics change. In other words, more data does not necessarily yield better estimates and, in fact, a longer-run Beta might turn out to be too backward-looking.¹ Regardless of the time period used, the Beta calculation is subject to both systematic error (we're using either too much or too little data) or sampling bias.

Several techniques have been implemented to adjust the Beta calculation for some of its drawbacks. The Blume and the Vasicek methods described below attempt to correct the sampling bias associated with the unadjusted Beta calculation, while the fundamental Beta calculation addresses the issue that a firm's Beta changes as its financial position evolves.

2 The Single-factor Model

A **single-factor** model for equity returns assumes that all return correlations arise from their common exposure to a single index, in this case that of the overall market. The overarching goal behind constructing this model is that it requires significantly less data to parameterize as opposed to the Markowitz model which requires having estimates of the correlation coefficient of any two stocks in an equity portfolio, in addition to their means and variances, in order to specify the return distribution of the portfolio.

The single-factor model is set up as follows. First, suppose that R_m is the return on the market index that is a synthetic portfolio in which all market assets are represented in the proportions they occur. For each stock return R_i , we set:

$$R_i = \alpha_i + \beta_i R_m + e_i, \tag{1}$$

where α_i and β_i are constants and the random variables e_i have mean zero and are uncorrelated to the market. We also make the crucial additional assumption that:

$$\mathbb{E}(e_i e_j) = 0, \text{ for } i \neq j;$$

that is to say, the variables e_i are not correlated with each other. This model assumption implies that the only reason why stocks vary together, systematically, is because of a common co-movement with the market. There are no effects beyond the market (e.g., industry effects) that account for co-movement between securities.

¹Cello currently uses two years of daily return history to estimate beta.

The Beta associated with each stock indicates its risk level compared to the market. For example, suppose that a stock has a Beta of 3 and an Alpha of 0. This means that if the market goes up by 1% the stock will have gone up by 3% on average.

Taken together, the assumptions we impose on the e_i allow us to deduce the following key theorem.

Theorem 1. *In a single-factor model:*

1. $\mathbb{E}(R_i) = \beta_i \mathbb{E}(R_m) + \alpha_i$;
2. $\text{Var}(R_i) = \beta_i^2 \sigma_m^2 + \sigma_{e_i}^2$;
3. The covariance between security i and security j is $\beta_i \beta_j \sigma_m^2$, for $i \neq j$.

Proof. The first result follows directly from the linearity of expectations and our assumption that $\mathbb{E}(e_i) = 0$. That is, we have:

$$\begin{aligned}\mathbb{E}(R_i) &= \mathbb{E}(\alpha_i + \beta_i R_m + e_i), \\ &= \alpha_i + \beta_i \mathbb{E}(R_m) + \mathbb{E}(e_i), \\ &= \alpha_i + \beta_i \mathbb{E}(R_m).\end{aligned}$$

The second result draws on the assumption that each e_i is uncorrelated with the market, and for uncorrelated random variables X and Y , we have:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

We therefore have:

$$\begin{aligned}\text{Var}(\alpha_i + \beta_i R_m + e_i) &= \text{Var}(\beta_i R_m) + \text{Var}(e_i), \\ &= \beta_i^2 \text{Var}(R_m) + \text{Var}(e_i).\end{aligned}$$

Note that since α_i is constant, it has no effect on variance.

Finally, to derive the expression for $\text{Cov}(R_i, R_j)$, we have from our definitions:

$$\begin{aligned}\text{Cov}(R_i, R_j) &= \mathbb{E}((R_i - \bar{R}_i)(R_j - \bar{R}_j)), \\ &= \mathbb{E}((\beta_i(R_m - \bar{R}_m) + e_i)(\beta_j(R_m - \bar{R}_m) + e_j)).\end{aligned}$$

Expanding and applying the linearity of expectation, this reduces to the sum of four terms. There are:

$$\begin{aligned}&\beta_i \beta_j \mathbb{E}((R_m - \bar{R}_m)^2), \\ &\beta_i \mathbb{E}((R_m - \bar{R}_m)e_j), \\ &\beta_j \mathbb{E}((R_m - \bar{R}_m)e_i), \\ &\mathbb{E}(e_i e_j).\end{aligned}$$

However, for $i \neq j$, the last three terms are equal to zero from our assumption that the terms e_i are uncorrelated both with the market and each other. $\square$

Note that the expected return has two components: a unique part α_i and a market-related part $\beta_i \bar{R}_m$. Likewise, the single-factor model divides the risk for the security into two pieces:

- The first part $\beta_i^2 \sigma_m^2$ arises from exposure to the market and is called **systematic risk**
- The remaining part $\sigma_{e_i}^2$ is called **specific risk**. It is the part unique to the security and it can be diversified away. Specific risk is also referred to as **alpha risk**, **diversifiable risk**, **unsystematic risk**, and **residual risk**.

In contrast, the covariance $\text{Cov}(R_i, R_j)$ depends only on market risk. This is what we meant earlier when we said that the single-factor model implied that the only reason securities move together is a common response to market movements.

Applying the single-factor to the market we have that:

$$R_m = 0 + 1 \cdot R_m,$$

which tells us that the market portfolio has $\alpha = 0$ and $\beta = 1$.

To gain a deeper understanding of why the β of a security is a measure of its systematic risk, we will show that by holding a sufficiently large number of securities we can effectively eliminate our exposure to residual risk. It follows that since we can diversify away this risk, we should not expect to obtain any risk premium for holding it. Thus, it is only the β that leads to a risk premium: we only receive a reward for our exposure to the overall level of the market.

We start by considering a portfolio of N assets, with weights X_i such that $\sum_{i=1}^n X_i = 1$. The portfolio risk may be computed as follows:

$$\begin{aligned} \text{Var}\left(\sum_{i=1}^n X_i R_i\right) &= \mathbb{E}\left(\left(\sum_{i=1}^n X_i (R_i - \bar{R}_i)\right)^2\right), \\ &= \mathbb{E}\left(\left(\sum_{i=1}^n X_i \beta_i (R_m - \bar{R}_m) + \sum_{i=1}^n X_i e_i\right)^2\right), \\ &= \mathbb{E}\left(\left(\sum_{i=1}^n \beta_i X_i\right)^2 (R_m - \bar{R}_m)^2\right) + \sum_{i=1}^n X_i^2 \mathbb{E}(e_i^2), \\ &= \left(\sum_{i=1}^n X_i \beta_i\right)^2 \sigma_m^2 + \sum_{i=1}^n X_i^2 \sigma_{e_i}^2. \end{aligned}$$

The Beta of the portfolio will be

$$\sum_{i=1}^n X_i \beta_i,$$

which is the weighted average of the Betas of the constituent stocks.

Suppose then that we divide our portfolio into a large number, N , of assets with equal amounts of money in each. Substituting $X_i = \frac{1}{N}$ into the above expression, we have that the portfolio variance is:

$$\sigma_P^2 = \beta_P^2 \sigma_m^2 + \frac{1}{N} \overline{\sigma_{e_i}^2}, \quad (2)$$

where the term β_P is the average of the stock Betas, and $\overline{\sigma_{e_i}^2}$ is the average of the idiosyncratic variances.

Now, provided that the individual risks $\sigma_{e_i}^2$ are bounded, the second term $\frac{1}{N} \overline{\sigma_{e_i}^2}$ will go to zero as N gets larger. It follows that by using a large portfolio of many different stocks, one can make the diversifiable risk disappear. In other words, the residual or specific risk for each of the stocks is diversified away.

Still, some risk does remain and that is the Beta or undiversifiable risk. We would therefore expect to receive risk premia for taking Beta risk but not diversifiable risk. We further note from equation (2) that the remaining residual risk is proportional to $1/N$. Hence even with as few as 10 stocks the residual risk is reduced to 10% of the average for the individual stocks. For a portfolio of 100 stocks, the residual risk is only 1% of the average.

3 Beta Estimation

Using a single-factor model implies that we need Beta estimates for each stock that we might potentially want to include in a portfolio. Estimates of future Betas could be arrived at by estimating Beta from past data and using this historical beta as an estimate of the future Beta. In addition to this technique, some of the other methods that have been proposed for estimating Beta include correcting historical Betas for the tendency of historical Betas to “regress to the mean,” and correcting historical estimates by incorporating fundamental firm data. We now discuss each of these estimation techniques.

3.1 Historical Betas

In Equation (1) we represented the return on a stock as:

$$R_i = \alpha_i + \beta_i R_m + e_i.$$

This equation is expected to hold at each moment in time although the values of α_i , β_i , or $\sigma_{e_i}^2$ might differ over time. While looking at historical data, one cannot directly observe α_i , β_i , or $\sigma_{e_i}^2$. Rather, one observes the past returns on the security and the market. If α_i , β_i , and $\sigma_{e_i}^2$ are assumed to be constant over time, then the same equation is expected to hold at each point in time and a straightforward regression analysis procedure exists for estimating α_i , β_i , or $\sigma_{e_i}^2$ over any time interval $t = 1, \dots, T$. These estimates are denoted by $\hat{\alpha}_i$, $\hat{\beta}_i$, and $\hat{\sigma}_{e_i}^2$.²

The regression approach shows that β_i is given by:

$$\beta_i = \frac{\text{Cov}(R_{i,t}, R_{m,t})}{\text{Var}(R_{m,t})}$$

The easiest way to compute α_i is first to compute β_i , and then, using

$$\mathbb{E}(R_i - (\alpha_i + \beta_i R_m)) = 0,$$

we have

$$\alpha_i = \mathbb{E}(R_i - \beta_i R_m) = \frac{1}{T} \sum_{t=1}^T (R_{i,t} - \beta_i R_{m,t}).$$

The $\hat{\alpha}_i$ and $\hat{\beta}_i$ produced by regression analysis are estimates of the true α_i and β_i that exist for a stock. The estimates are subject to error. Furthermore, the estimation process is also hamstrung by the fact that α_i and β_i are not perfectly stationary over time. We would expect changes in Beta as the fundamental characteristics of the firm change. For example, β_i as a risk measure should be related to the capital structure of the firm and, thus, should change as the capital structure changes.

3.2 Accuracy of Historical Betas

The first logical step in looking at Betas is to see how much association there is between the Betas in one period and the Betas in an adjacent period. A study by Blume [Blume70] computes Betas using time series regressions on monthly data for nonoverlapping seven-year periods. Betas are generated for single stock portfolios, 2 stock portfolios, 4 stock portfolios, and so forth up to 50 stock portfolios, and for each portfolio it is examined how highly correlated Betas from one period were with the Betas for a second period. The correlations between periods increase dramatically as the number of stocks go up. Changes in security Betas

²We shall generally use the “hat” notation to denote any estimate of α_i , β_i , or $\sigma_{e_i}^2$, whether obtained through regression analysis or some other technique.

will differ from security to security. These results intuitively make sense as one would expect that the errors in estimating Betas for individual securities would tend to cancel out when securities are combined, and therefore, there would be less error in measuring a portfolio's Beta.

3.3 Adjusting Historical Betas

To attempt to further improve the predictive ability of Betas on securities and portfolios we examine a simple hypothetical distribution of Betas. The Beta we calculate for any stock will be, in part, a function of truly underlying Beta and, in part, a function of sampling error. If we compute a very high estimate of Beta for a stock, we have an increased probability that we have positive sampling error, while if we compute a very low estimate of Beta, we have an increased chance that we have a negative sampling error. If this line of argument is correct we should find that Betas, on average, tend to converge to 1 in successive time periods. Estimated Betas that are lot larger than one should tend to be followed by estimated Betas that are closer to one (lower), and estimated Betas below one should tend to be followed by higher Betas. Evidence that this does happen has been presented by Blume [Blume75].

Blume's Technique

Suppose that we measure Betas of N stocks in one period, $\hat{\beta}_{i,1}$, and then remeasure them for the same stocks in the next period $\hat{\beta}_{i,2}$ with $i = 1, 2, \dots, N$. If we're correct about the tendency of Betas to regress towards a central value then the high Betas should tend to go down and the low Betas should tend to go up. We can use regression to test this hypothesis. We set up the regression equation:

$$\hat{\beta}_{i,2} \sim a\hat{\beta}_{i,1} + b$$

where i ranges over all the stocks and a and b are the same for all the stocks. Now, if our hypothesis holds, we should get $a < 1$.

To consider a concrete example suppose that we are working with a fund that has a 5-year time horizon. We want to make the best estimate we can of the Beta of each stock for the next 5 years and adopt the following strategy. We regress Betas from 0 to 5 years ago, $\hat{\beta}_i^{\text{last}}$, against ones from 5 to 10 years ago, $\hat{\beta}_i^{\text{previous}}$, to get a, b so that:

$$\hat{\beta}_i^{\text{last}} = a\hat{\beta}_i^{\text{previous}} + b,$$

is as well-fitted as possible.

Once we have a and b , we can then use β for the last 5 years for each stock, $\hat{\beta}_i^{\text{last}}$, and we put:

$$\hat{\beta}_i^{\text{next}} = a\hat{\beta}_i^{\text{last}} + b,$$

to estimate for the next 5 years. Note that if $a + b = 1$, we can write $w_1 = a, w_2 = b$ and rewrite Blume's equation as:

$$\hat{\beta}_i^{\text{next}} = w_1 \cdot \hat{\beta}_i^{\text{last}} + w_2 \cdot 1.$$

When viewed from this lens, we can view Blume's technique as adjusting the historical Beta estimate towards 1, based on the relative weights of a and b .

Blume's method [Blume71] is somewhat *ad hoc* and does not have a firm mathematical foundation but has value as a generally plausible idea. It does not take into account that for stocks with low idiosyncratic risk, the beta will be more accurate, which would suggest that less scaling is required for such stocks. Another issue is that it does not ensure that a portfolio with $\beta = 1$ and no idiosyncratic risk, i.e., the market, is mapped to a portfolio with $\beta = 1$.

Vasicek's Technique

It would be desirable not to adjust all stocks the same amount towards the average but rather to have the adjustment depend on the size of the uncertainty (sampling error) about Beta. The larger the sampling error, the greater the chance of large differences from the average being due to sampling error and the greater the adjustment. Vasicek [Vasicek73] suggests a scheme that incorporates these properties. If we let $\text{avg}(\hat{\beta}_1)$ equal the average Beta, across the sample of stocks in the historical period, then the Vasicek procedure involves taking a weighted average of an estimator involving $\text{avg}(\hat{\beta}_1)$ and the historical Beta for security i . Specifically, let $\hat{\sigma}_{\text{avg}(\hat{\beta}_1)}^2$ denote the variance of the distribution of the historical estimates of Beta over the sample of stocks. Also, let $\hat{\sigma}_{i,1}^2$ stand for the square of the standard error estimate of Beta for security i measured in time period 1. This is a measure of the uncertainty associated with the measurement of the individual security's Beta. Then, the forecast of Beta for security i is:

$$\hat{\beta}_{i,2} = \frac{\hat{\sigma}_{i,1}^2}{\hat{\sigma}_{\text{avg}(\hat{\beta}_1)}^2 + \hat{\sigma}_{i,1}^2} \text{avg}(\hat{\beta}_1) + \frac{\hat{\sigma}_{\text{avg}(\hat{\beta}_1)}^2}{\hat{\sigma}_{\text{avg}(\hat{\beta}_1)}^2 + \hat{\sigma}_{i,1}^2} \hat{\beta}_{i,1}$$

Fundamental Betas

Beta is a risk measure that arises from the relationship between the return on a stock and the return on the market. However, we know that the risk of a firm should be determined by some combination of the firm's fundamentals. For example, some relationships we would expect to see:

- Growth is usually thought of as positively associated with Beta. High-growth firms are thought of as more risky than low-growth firms.
- Leverage tends to increase the volatility of the earnings stream, hence to increase risk and Beta.
- Large firms are often thought to be less risky than small firms, if for no other reason than that they have better access to the capital markets. Hence, they should have lower Betas.

A fundamental Beta model relates Beta to several fundamental variables via multiple regression analysis. In practice, the approach taken is to combine both fundamental and historical Betas in one system.

4 Implementing Beta Adjustments

In this analysis, we set up a horse race in which we see which of the three Beta calculations: Unadjusted, Blume and Vasicek, does the best job of predicting the "realized" Beta for the next period. Since the basic "unadjusted" beta calculation relies on 2 years of history, each period is set up to have 500 trading dates. The analysis is based on daily returns from January 2003 to January 2015 on a sample of 65 real-estate linked stocks. We subdivide this 12 year period into two 6-year intervals, Interval I and Interval II. Each Interval is then further broken into three 2-year periods. The first two periods in the interval are used to come up with a Beta prediction³ for the next period. We compute the mean-squared-error (MSE) of this forecast relative to the realized Beta for the third period. We compute Betas with respect to both the S&P 500 and the IYR for the purposes of this exercise.

The results are summarized in Table 1 and 2. When the Betas are calculated with respect to the stock market, Blume's technique noticeably outperforms both the Unadjusted and Vasicek techniques. When the Beta is with respect to the IYR, the results from the different techniques are broadly the same although both the Beta Adjustment techniques outperform the Unadjusted technique.

³Only the Blume technique needs the full 4 year period, the Unadjusted and Vasicek methods just rely on two years of data.

Method	Interval I	Interval II
Unadjusted	0.0728	0.0484
Blume	0.0454	0.0157
Vasicek	0.0724	0.0477

Table 1: MSE of Beta Estimates (S&P 500)

Method	Interval I	Interval II
Unadjusted	0.0297	0.0272
Blume	0.0296	0.0210
Vasicek	0.0295	0.0261

Table 2: MSE of Beta Estimates (IYR)

References

- [1] Blume, Marchall. April 1970. "Portfolio Theory: A Step Towards Its Practical Application," *Journal of Business*, **43**, No. 2, pp.152-173.
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- [3] Vasicek, Oldrich. December 1973. "A Note on Using Cross-Sectional Information in Bayesian Estimation of Security Betas," *Journal of Finance*, **VIII**, No. 5, pp. 1233-1239.